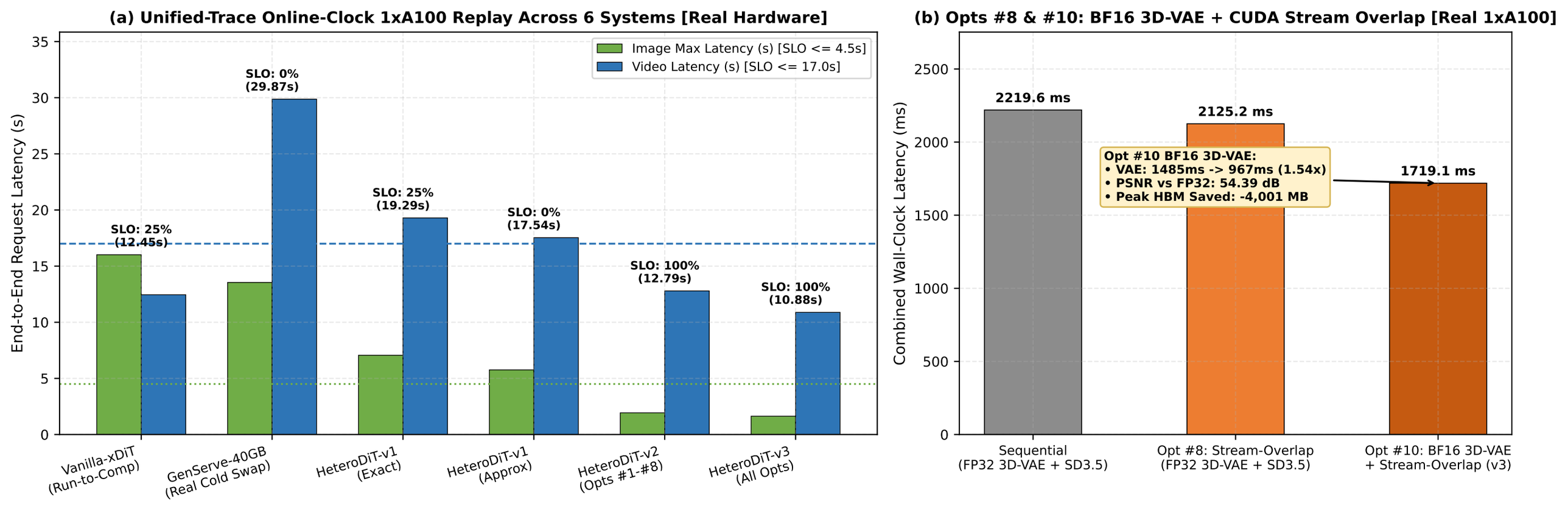
(a) Unified-Trace Online-Clock 1xA100 Replay Across 6 Systems [Real Hardware]
End-to-End Request Latency (s)
Image Max Latency (s) [SLO <= 4.5s]
Video Latency (s) [SLO <= 17.0s]
SLO: 25%
(12.45s)
SLO: 0%
(29.87s)
SLO: 25%
(19.29s)
SLO: 0%
(17.54s)
SLO: 100%
(12.79s)
SLO: 100%
(10.88s)
Vanilla-xDiT
(Run-to-Comp)
GenServe-40GB
(Real Cold Swap)
HeteroDiT-v1
(Exact)
HeteroDiT-v1
(Approx)
HeteroDiT-v2
(Opts #1-#8)
HeteroDiT-v3
(All Opts)
(b) Opts #8 & #10: BF16 3D-VAE + CUDA Stream Overlap [Real 1xA100]
Combined Wall-Clock Latency (ms)
2219.6 ms
2125.2 ms
1719.1 ms
Opt #10 BF16 3D-VAE:
• VAE: 1485ms -> 967ms (1.54x)
• PSNR vs FP32: 54.39 dB
• Peak HBM Saved: -4,001 MB
Sequential
(FP32 3D-VAE + SD3.5)
Opt #8: Stream-Overlap
(FP32 3D-VAE + SD3.5)
Opt #10: BF16 3D-VAE
+ Stream-Overlap (v3)